AI-Powered Research Gap Identifier & Trend Analyzer

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

Academic research project management involves several connected decisions, including research idea validation, supervisor matching, integrity checking, collaboration, progress monitoring and feedback. In many final-year and postgraduate research settings, these activities are handled through fragmented manual processes, static documents, informal communication and isolated administrative tools. This common integrated solution report presents ResearchLens, an AI-powered research lifecycle platform that brings four interdependent components into one cohesive workflow: semantic research gap detection, supervisor recommendation and collaboration, integrity governance with blockchain anchoring, and collaborative workspace support with NLP-assisted supervision.

The integrated solution uses a React-based user interface, a Node.js/Express gateway, PostgreSQL-compatible persistence, and specialised Python-based AI and governance services. The system supports idea-level semantic gap analysis, KeyBERT and graph-based supervisor recommendation, tamper-evident submission governance, logbook management, abstractive summarisation and supervisor feedback support. Scenario-based and integration-focused validation showed that the components can operate as a connected workflow rather than four isolated modules. The value of the complete solution is its ability to support students, supervisors and coordinators across the research lifecycle while preserving human review, role-based access and institutional control.

Keywords: Academic Research Management, Supervisor Recommendation, Research Gap Detection, Integrity Governance

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LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
AI	Artificial Intelligence
API	Application Programming Interface
BART	Bidirectional and Auto-Regressive Transformers
DAG	Directed Acyclic Graph
FAISS	Facebook AI Similarity Search
GNN	Graph Neural Network
JWT	JSON Web Token
LLM	Large Language Model
NLP	Natural Language Processing
RBAC	Role-Based Access Control
SDG	Sustainable Development Goal
TAF	Topic Assessment Form

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Research project management in higher education requires students, supervisors and coordinators to make decisions across the full research lifecycle. Students must identify viable research ideas, locate suitable supervisors, submit formal documents, maintain progress records and respond to academic feedback. Supervisors must review proposals, monitor progress, provide feedback and ensure that research outputs remain authentic and traceable. Coordinators must manage allocations, maintain records and uphold academic integrity across multiple research groups.

The common problem addressed by this project is that these activities are often distributed across separate documents, emails, spreadsheets, messaging tools and manually checked submissions. This fragmentation creates delays, inconsistent decision-making, limited visibility and weak traceability. The integrated solution responds to this context by combining AI-assisted analysis, workflow automation and integrity governance into a single academic research lifecycle platform.

1.2 Stakeholders and user needs

The primary stakeholders are students, academic supervisors, co-supervisors, project coordinators and academic administrators. Students need guidance when validating research ideas, selecting supervisors, submitting research artefacts and tracking progress. Supervisors need concise visibility of student activity, support for reviewing logbooks, access to relevant research context and reliable request management. Coordinators and administrators need oversight of groups, supervision allocation, academic integrity evidence and progress across the cohort.

The user needs identified across the four components are therefore connected rather than independent. Students need idea validation before committing to a topic, suitable supervisors before proceeding with project work, trusted submission handling before assessment and continuous collaboration support after supervision begins. Supervisors and coordinators need the same process to remain auditable, role-controlled and manageable at scale.

1.3 Alignment with the Sustainable Development Goals

The project is most closely aligned with SDG 4: Quality Education because it aims to improve the structure, transparency and support available during academic research supervision. By assisting research idea validation, supervisor matching, progress tracking and feedback, the platform supports a more informed and consistent student research experience.

The solution also has relevance to SDG 9: Industry, Innovation and Infrastructure because it demonstrates how responsible digital infrastructure, AI-assisted services and secure workflow design can be applied within educational institutions. The contribution is framed as decision support and academic process improvement rather than a claim of direct societal impact.

1.4 Purpose and scope of the common report

This common report presents the integrated project-level problem, solution structure, architecture, component relationships, integration approach and overall validation of the complete system. It does not duplicate the four individual theses. Instead, it explains how the components work together as one platform and how the shared architecture supports the end-to-end research lifecycle.

The detailed algorithms, datasets, component-level experiments and individual implementation evidence are documented in the four individual theses. This report focuses on the common value of the complete solution, shared interfaces, coordinated workflows, group-level objectives, integrated validation and sustainability considerations.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The overall problem is that academic research project workflows are frequently managed through fragmented and partially manual processes that do not adequately connect research idea validation, supervisor allocation, integrity verification and ongoing supervision support. As a result, students may pursue topics without sufficient evidence of novelty, approach unsuitable or overloaded supervisors, submit documents without strong tamper-evident traceability, and maintain progress records that supervisors cannot efficiently review.

The problem is not simply the absence of a single tool. It is the lack of an integrated workflow that supports evidence-based academic decision-making from the point where a student proposes an idea to the point where supervisors review progress and institutional records are preserved.

2.2 Root causes and overall research gap

The root causes include reliance on manual literature checking, static staff profile browsing, informal supervisor recommendations, separated document submission processes and delayed review of student progress. Existing tools often address only one part of the process, such as literature discovery, project management, plagiarism checking or messaging. They do not sufficiently connect idea validation, supervisor selection, integrity governance and supervisor feedback into a unified academic research workflow.

The overall research gap is therefore the absence of a cohesive platform-level approach that combines semantic analysis, recommendation, governance and collaboration while preserving human oversight. The project addresses this gap through four interdependent components that share common users, data flows, authentication, storage and operational workflows.

2.3 Subproblem and component mapping

The project separates the overall problem into four non-overlapping but interdependent subproblems. Each component addresses a specific stage of the

research lifecycle, but the components are designed to exchange data through the shared platform rather than operate as separate products.

Table 1 . summarises the mapping between subproblems, components and their contribution to the integrated solution.

Subproblem	Component	Owner	Role in overall solution
Early-stage research idea validation	Semantic Analysis and Gap Detection Module	S.M.D.M. Bandara	Retrieves and analyses literature evidence to classify whether a student idea addresses a gap, partially overlaps or is highly similar to existing work.
Supervisor discovery and allocation	Recommendation and Collaboration Network	K.L. Widanagama	Matches research ideas to suitable supervisors using topic understanding, graph-based relevance and workload feasibility, then supports request and assignment workflow.
Academic integrity and provenance	Integrity Governance and Blockchain Ledger	Darshan R	Creates tamper-evident academic event records using a local DAG ledger, hashing, duplicate detection and blockchain anchoring.
Progress monitoring and feedback	Collaborative Workspace and NLP Supervisor Assistant	Wijesinghe L.P.P.	Supports logbooks, collaboration, progress tracking, AI-assisted summarisation and supervisor feedback

			support.

2.4 Overall aim and group objectives

The common aim of the project is to design and implement an integrated AI-powered academic research lifecycle platform that assists students, supervisors and coordinators from research idea validation to supervisor allocation, integrity verification and progress monitoring.

Table 2 . The group objectives, mapped component and the type of evidence used for evaluation

Objective	Mapped component(s)	Evidence used for evaluation
Support evidence-based research idea validation before topic commitment.	Semantic Gap Detection Module	Scenario tests, classification output, evidence retrieval review and processing time observations.
Recommend supervisors using topic relevance, relationship awareness and workload feasibility.	Recommendation and Collaboration Network	Recommendation output, request workflow tests, workload display and assignment state validation.
Provide tamper-evident integrity governance for research submissions and academic events.	Integrity Governance and Blockchain Ledger	Hash validation, ledger verification, duplicate detection and attestation status tests.
Support ongoing collaboration, logbook review and supervisor feedback.	Collaborative Workspace and NLP Supervisor Assistant	Logbook workflow testing, summarisation behaviour, supervisor review and communication scenarios.
Integrate the four components into one controlled research lifecycle platform.	All components	API integration tests, database state checks, role-based access behaviour and end-to-end scenario walkthroughs.

2.5 Scope assumptions and constraints

The scope of the complete solution covers academic research idea assessment, supervisor recommendation and request handling, integrity verification of selected

submissions, logbook support and supervisor review assistance. The solution assumes that users are authenticated within an institutional environment and that student groups, supervisor profiles and relevant academic records are available through the platform database.

The project does not claim to replace academic judgement, supervisor approval, institutional policies or final assessment decisions. AI-generated outputs are used as decision-support artefacts. The solution also depends on the quality of input data, the availability of AI services, the correctness of shared identifiers and the security of the hosting environment.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

ResearchLens is designed as a distributed academic research lifecycle platform. Its central value proposition is to connect stages that are usually handled separately: idea exploration, supervisor recommendation, formal request handling, submission integrity checking, logbook management and supervisor feedback support. The platform therefore treats academic research supervision as a continuous workflow instead of a collection of isolated tasks.

The integrated outcome is a student and supervisor journey in which a student can test a proposed idea against research evidence, receive suitable supervisor recommendations, send role-specific requests, submit project documents into a tamper-evident workflow, maintain research logbooks and receive structured supervisor feedback. Coordinators benefit from better visibility, consistency and traceability across the process.

3.2 High level architecture

The high-level architecture follows a multi-tier design. The client tier contains React-based student, supervisor and administrative interfaces. The application gateway tier uses a Node.js/Express backend to handle authentication, role-based access control, validation, routing and transactional business rules. The specialised service tier contains AI and governance services for semantic gap detection, recommendation, NLP summarisation and integrity anchoring. The data tier provides shared relational storage and document storage.

This architecture allows common workflow operations to remain under the control of the central gateway while specialised AI workloads are handled by independent Python services. Figure 1 shows the main deployment and integration boundaries.

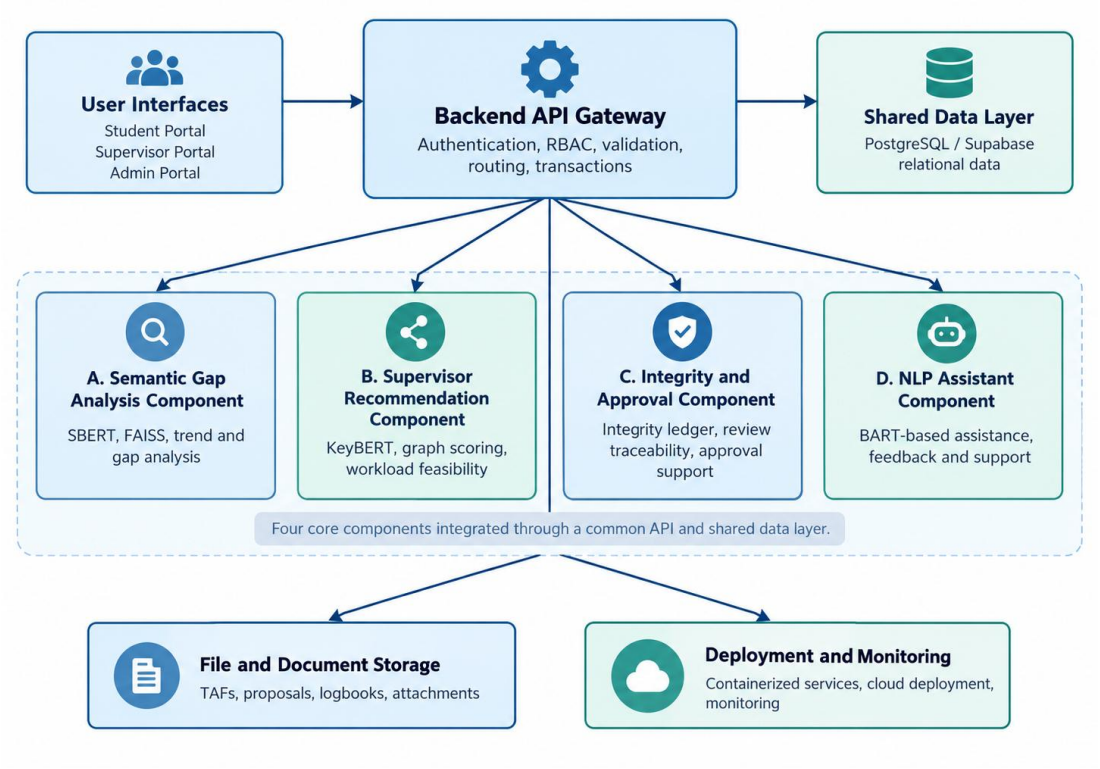


Figure 1. High-level integrated architecture

3.3 Component 1 summary

The Semantic Analysis and Gap Detection Module addresses the problem of early-stage research idea validation. Its input is a student research idea or proposal text. The component retrieves and analyses relevant literature, represents text using dense semantic embeddings, extracts gap-related evidence and uses a reasoning layer to classify the idea as gap-addressing, partially overlapping or highly similar to existing work.

Its output is an evidence-supported judgement that helps students and supervisors understand whether the idea appears novel enough to proceed or requires refinement. This component contributes to the overall solution by improving the quality of topics before supervisor matching and formal project work begin.

3.4 Component 2 summary

The Recommendation and Collaboration Network addresses the problem of supervisor discovery and allocation. Its input is the student research title, abstract and keywords. The component uses KeyBERT-based keyword extraction, topic

similarity, graph-based academic relationship scoring and workload feasibility to generate ranked supervisor recommendations.

Its output is a set of recommendation cards that support role-specific supervisor or co-supervisor requests. This component contributes to the overall solution by connecting student research ideas with relevant academic staff while considering practical supervision capacity and preserving supervisor approval.

3.5 Component 3 summary

The Integrity Governance and Blockchain Ledger component addresses the problem of preserving trust in academic events and submissions. Its input is a formal academic event or document submission, such as a Topic Assessment Form or proposal artefact. The component creates cryptographic hashes, maintains a local DAG-based provenance ledger, detects duplicates or similar submissions and anchors proof information on a blockchain network without placing private academic content on-chain.

Its output is an integrity status that supports tamper detection, duplicate checking and auditability. This component contributes to the overall solution by making academic records more trustworthy while keeping the operational workflow responsive for users.

3.6 Component 4 summary

The Collaborative Workspace and NLP Supervisor Assistant addresses the problem of ongoing supervision, communication and progress monitoring. Its input includes student logbook entries, activity descriptions, tasks and group communication records. The component manages logbook submission, supervisor review, real-time collaboration and AI-assisted summarisation of student activities.

Its output includes structured logbook records, concise summaries, supervisor feedback support and progress visibility. This component contributes to the overall solution by reducing feedback latency and helping supervisors review research progress more efficiently while retaining access to the original student submissions.

3.7 End to end user and data flow

The end-to-end workflow begins when a student enters or submits a research idea. The idea can be analysed for semantic gap evidence, then used to obtain supervisor recommendations. After the student sends a supervisor or co-supervisor request, the supervisor reviews the request and either accepts or rejects it. Accepted assignments become part of the group record and influence later workload calculations.

As the project progresses, formal submissions can be processed through the integrity workflow, while logbook activity is captured in the collaborative workspace. Supervisors can review logbook entries with the support of summarisation and AI-assisted feedback suggestions. Figure 2 summarises this user and data flow across the four components.

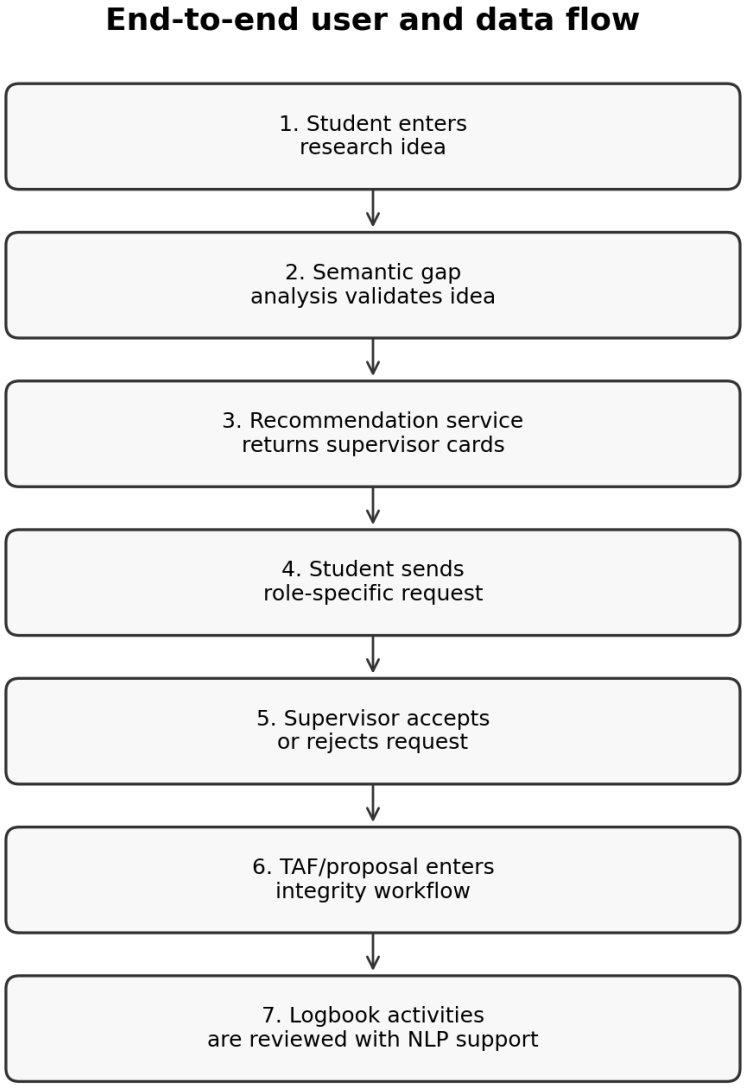


Figure 2 . End-to-end user and data flow

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The integration strategy follows an API gateway and microservice choreography pattern. The Node.js/Express backend acts as the public-facing gateway that receives user requests, enforces authentication, validates payloads and coordinates database transactions. Specialised Python services are invoked for model inference, semantic analysis, recommendation, summarisation and integrity processing.

This approach was selected because it separates general application workflow from specialised AI and governance workloads. It also reduces direct exposure of internal services, allows components to be developed independently and supports future scaling of services that require heavier processing.

4.2 Interfaces data contracts and shared models

The platform depends on shared data contracts so that components can exchange outputs without breaking downstream workflows. The central gateway standardises requests and responses, while shared records represent users, groups, supervisor requests, submissions, logbooks, integrity events and component outputs. API contracts define the fields required by each service, expected response format and error-handling responsibility.

Table 3 . summarises the main shared interfaces used by the integrated solution.

Interface or model	Primary producer	Primary consumer	Purpose
Student research idea input	React frontend / Node gateway	Semantic gap service and recommendation service	Transfers title, abstract, keywords and related metadata for analysis and recommendation.
Supervisor recommendation response	Recommendation service	Node gateway and student interface	Returns ranked supervisor cards with relevance, feasibility and availability information.

Supervision request record	Node gateway	Supervisor dashboard and database workflow	Stores role-specific requests, status, group details and supervisor decisions.
TAF or proposal submission event	Node gateway	Integrity ledger service	Provides document hashes and metadata for duplicate detection and attestation.
Logbook review context	Node gateway	NLP summarisation service and supervisor interface	Transfers student activity text for summarisation and feedback support.

4.3 Shared nonfunctional requirements

The complete platform requires performance, reliability, security, privacy, maintainability and usability across all components. Performance is important because recommendation results, summaries and validation feedback should be returned within an acceptable interactive window. Reliability is important because academic workflows should not leave partially completed assignments, submissions or approval states.

Security and privacy are addressed through authentication, role-based access control, input validation and controlled access to student and supervisor records. Maintainability is supported by modular service boundaries, shared API contracts and version-controlled implementation artefacts. Responsible AI requirements apply across the platform because recommendations, classifications and summaries should support human judgement rather than replace it.

4.4 Integrated workflow and exception handling

The integrated workflow includes several exception paths that must be handled safely. If a specialised AI service is temporarily unavailable, the gateway should return a clear error or apply a defined fallback rather than blocking unrelated platform functions. If two supervisors attempt to accept conflicting requests, the

assignment workflow must use safe transactional handling to prevent inconsistent role allocation.

Document submission and integrity processing are handled so that user-facing submission can complete while slower attestation and verification activities proceed asynchronously. Logbook summarisation and feedback support should also preserve access to the original logbook entry so that supervisors can verify the AI-generated summary before using it.

4.5 Team contribution and dependency matrix

The project required a deliberate division of work because each component addresses a different stage of the research lifecycle. At the same time, the components are mutually dependent through shared users, groups, documents, supervisor records and workflow states.

Table 4 . summarises the main team responsibilities and integration dependencies.

Team member	**Main component**	**Key individual contribution**	**Shared integration contribution**
S.M.D.M. Bandara	Semantic Analysis and Gap Detection	Designed the idea-level semantic gap detection pipeline and output interpretation.	Connected gap evidence to the wider student research lifecycle.
K.L. Widanagama	Recommendation and Collaboration Network	Developed topic-aware, graph-aware and workload-aware supervisor recommendation and request workflow.	Integrated supervisor recommendation outputs with group assignment and workload state.
Darshan R	Integrity Governance and Blockchain Ledger	Implemented DAG-based provenance, hashing, duplicate detection and blockchain anchoring.	Integrated submission events and attestation status with the core workflow.
Wijesinghe L.P.P.	Collaborative Workspace and NLP	Developed logbook workflow,	Integrated supervision progress data, chat and

	Supervisor Assistant	collaboration features and BART-based summarisation support.	feedback support with supervisor dashboard functions.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The common frontend technology stack uses React with Vite to provide student, supervisor and administrative interfaces. The shared backend uses Node.js and Express to provide authentication, routing, validation, API orchestration, role-based access and database communication. PostgreSQL-compatible storage is used for structured records, while file or document storage is used for submissions and attachments.

The AI and governance services are implemented using Python-based frameworks such as FastAPI or Flask depending on the component. The model and analysis stack includes SentenceTransformers, FAISS, KeyBERT, PyTorch Geometric, BART-based summarisation and related NLP tooling. Docker-based or cloud-ready deployment is used to separate service environments and simplify future operational deployment.

5.2 Development and integration process

Development was organised around component ownership, shared data models and iterative integration. Component teams first developed core functionality within their own services, then connected their outputs to the central backend and shared database. Interface-first coordination was necessary because the output of one component often became the input or state condition for another component.

Integration testing was carried out through scenario walkthroughs that represented realistic student and supervisor journeys. These scenarios were used to confirm that API contracts, role-based access, database updates and user interface states remained consistent across the combined platform.

5.3 Deployment and operational configuration

The integrated solution can be deployed using a cloud-hosted architecture in which the frontend, Node.js gateway, Python microservices and PostgreSQL-compatible database are configured as separate services. Microsoft Azure can be used for

deployment through services such as Azure App Service or Azure Container Apps for the web and microservice layers, with Azure Database for PostgreSQL or another PostgreSQL-compatible managed service for persistence.

Operational configuration requires secure environment variables, database credentials, service URLs, API keys, storage configuration and logging settings. Monitoring should cover API failures, model service availability, database errors, attestation queue status and user workflow failures. Backups and migration scripts are required to preserve academic records and allow controlled recovery.

5.4 Evidence of team execution

Evidence of team execution includes the completed component implementations, integration scenarios, API tests, database state checks, user interface walkthroughs and validation notes documented across the individual theses. The common report summarises this evidence at group level rather than reproducing raw logs or source code.

The most important execution evidence is that the components can participate in a connected workflow: research ideas can be analysed, recommended supervisors can be requested, assignment state can be updated, submissions can enter integrity processing and logbook activity can be reviewed by supervisors.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The group-level evaluation focused on whether the platform works as an integrated research lifecycle solution. Success criteria included correct data flow between components, valid API communication, appropriate role-based access, consistent database updates, understandable user interface behaviour and the ability to complete representative end-to-end scenarios.

Each component also used component-specific validation methods in its individual thesis. At the common-report level, the evaluation emphasises integration, workflow completeness, evidence of cooperation between services and the practical usefulness of the overall solution.

6.2 Integration and end to end testing

End-to-end testing used scenario-based walkthroughs that follow the normal academic workflow from research idea input to supervision and progress review. The purpose was to confirm that the complete platform behaves consistently when multiple components operate through shared records and service calls.

Table 5 . summarises representative integrated test areas and the type of evidence used to verify behaviour.

Test area	Scenario	Expected integrated behaviour	Evidence type
Idea validation to recommendation	Student enters a research idea and requests analysis or supervisor matching.	The system processes the idea and returns gap evidence or suitable supervisor recommendations.	Service response, UI output and database records.
Supervisor request and assignment	Student sends a role-specific request and supervisor accepts it.	The correct role is assigned and conflicting pending requests are handled safely.	API response, database state and dashboard review.
Document integrity	Student submits a	Submission is stored,	Hash record, queue

workflow	TAF or proposal document.	hashed and queued or processed for integrity verification.	state and attestation status.
Logbook review workflow	Student submits progress activities and supervisor opens review view.	The logbook is visible with AI-assisted summary or fallback review information.	UI state, summary response and approval record.
Role-based access	Student, supervisor and admin access different functions.	Users only perform operations permitted for their role.	Endpoint behaviour and interface restrictions.

6.3 Overall results

The overall result is a working integrated prototype that demonstrates how semantic gap detection, supervisor recommendation, integrity governance and collaborative supervision support can operate within one platform. The system supports structured student input, service-based AI analysis, shared database state, role-specific workflow operations and supervisor-facing review actions.

The results also show that the complete solution provides value beyond the individual components. A ranked supervisor recommendation becomes more useful when it follows idea validation and connects to request handling. Integrity verification becomes more meaningful when it is tied to actual academic submissions. Logbook summarisation becomes more useful when it is connected to the group, supervisor and research workflow already stored in the platform.

6.4 Stakeholder and user validation

Stakeholder validation at the common-report level is represented through scenario-based review rather than full institutional deployment. The scenarios reflect realistic student, supervisor and coordinator interactions, including entering research ideas, reviewing recommendations, sending requests, checking integrity status, submitting logbooks and reviewing progress.

The validation considered whether the workflow is understandable, whether required information appears at the correct stage, and whether human review remains in

control of important academic decisions. The scenarios indicated that the platform structure is aligned with institutional supervision practice because students initiate actions, supervisors review and decide, and coordinators can rely on stored workflow evidence.

6.5 Achievement of common objectives

The common objectives were achieved at prototype level. The platform demonstrates the intended flow across the four components, although larger-scale deployment, institutional data migration and extended stakeholder trials remain future work. The achievement status is summarised below.

Table 6 . common objective,status and justification

Common objective	**Status**	**Justification**
Support evidence-based research idea validation.	Achieved in prototype	Semantic analysis and gap classification workflow implemented and tested through scenarios.
Support topic-aware and workload-aware supervisor matching.	Achieved in prototype	Recommendation cards, workload visibility, request creation and assignment workflow implemented.
Support integrity governance for submissions and academic events.	Achieved in prototype	Hashing, provenance ledger, duplicate checking and attestation workflow implemented.
Support collaboration, logbook review and AI-assisted feedback.	Achieved in prototype	Logbook workflow, summarisation support and supervisor-facing review actions implemented.
Integrate components as a complete platform.	Partly achieved / prototype level	Core integration was validated in controlled scenarios; wider institutional deployment and larger-scale testing remain future work.

6.6 Integrated limitations and threats

The integrated solution has several limitations. The prototype depends on the accuracy of supervisor profiles, research topic metadata, document text extraction, logbook quality and configuration of each service. AI model outputs may be affected by input quality, limited datasets or incomplete contextual information. Performance may vary when transformer-based services experience cold starts or when multiple students use services simultaneously during registration periods.

System-level threats include inconsistent shared identifiers, stale workload data, service outages, prompt injection attempts in free-text fields, oversized document uploads and concurrency issues during supervisor acceptance. These risks are mitigated through validation, role-based access, service separation, fallback behaviour, database transactions and future monitoring, but they require careful management before production deployment.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The value proposition of the integrated solution is to reduce uncertainty, delay and fragmentation in academic research project management. Students benefit from earlier idea validation, clearer supervisor guidance, structured requests and better progress support. Supervisors benefit from more organised requests, concise progress summaries, integrity visibility and a clearer review workflow.

Project coordinators and departments benefit because the platform improves visibility across groups, supervision allocation, submissions and progress evidence. The wider institutional benefit is a more structured and auditable research lifecycle that combines intelligent support with human academic decision-making.

7.2 Market alternatives and differentiation

Existing alternatives include literature search tools, plagiarism checkers, learning management systems, staff profile pages, project management boards and messaging platforms. These tools may support parts of the research workflow, but they are usually not designed as one integrated academic research lifecycle system.

The platform is differentiated by combining idea-level semantic gap analysis, supervisor recommendation, integrity governance and collaborative supervision support in one workflow. Its competitive position is therefore not as a generic recommender or chatbot tool, but as an academic workflow enhancement for institutions managing research groups and supervisor allocation.

7.3 Business model and feasibility

A sustainable model can be based on institutional deployment for universities, faculties or departments that manage undergraduate and postgraduate research projects. The platform can be configured according to institutional roles, workflows, supervision rules, capacity thresholds and assessment documentation requirements.

Operational feasibility depends on maintaining the shared database, service configuration, AI model artefacts, deployment pipeline, documentation and support process. A department-level or institutional subscription model, internal deployment model or managed-service model could be considered depending on data protection and governance requirements.

7.4 Adoption intellectual property and compliance

Adoption requires institutional confidence in data privacy, role-based access, reliability and responsible AI use. Intellectual property considerations include ownership of source code, model configuration, system documentation, generated diagrams, synthetic datasets and institutional data. External libraries and APIs must be used according to their licences and service terms.

Compliance considerations include protection of student and supervisor data, access control, retention of academic records, auditability of submissions and transparency of AI-assisted outputs. The platform should make clear that AI-generated recommendations, classifications and summaries are decision-support outputs rather than automatic final decisions.

7.5 Sustainable Development Goal contribution

The project contributes most directly to SDG 4: Quality Education by improving support for student research, supervisor allocation, progress monitoring and academic integrity. It may also support SDG 9: Industry, Innovation and Infrastructure by demonstrating responsible use of AI, secure digital workflows and cloud-ready service architecture within educational infrastructure.

To sustain these benefits, the platform must be maintained with accurate data, transparent governance, clear academic policies and continued human oversight. Possible negative effects, such as over-reliance on AI outputs or unfair outcomes caused by poor data quality, should be managed through explainability, review procedures and institutional accountability.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The integrated project addresses the common problem of fragmented academic research management by providing a connected platform for idea validation, supervisor recommendation, integrity governance and collaborative supervision support. The solution demonstrates that research lifecycle management can be improved when AI-assisted analysis and workflow automation are integrated with role-based human review.

The evidence from component implementation and integration scenarios indicates that the platform can support students, supervisors and coordinators across key stages of the academic research process. The project therefore provides a practical prototype foundation for more transparent, efficient and accountable research supervision.

8.2 Principal integrated contribution

The principal integrated contribution is the design and implementation of a cohesive academic research lifecycle platform that connects four complementary research components through shared users, data, APIs and workflows. The contribution is not limited to separate algorithms; it is the coordination of semantic analysis, recommendation, governance and collaboration into one operational system.

This integrated contribution shows that academic research support systems should be designed around the full user journey. A research idea, supervisor match, submission record and progress log are connected academic artefacts, and the platform treats them as part of a common lifecycle rather than isolated records.

8.3 Future directions

Future work should focus on institutional pilot deployment with real users, larger and more diverse datasets, stronger performance testing and improved monitoring. Additional work is also needed to refine model explainability, expand supervisor and research-area coverage, strengthen data governance and validate the system with coordinators and academic panels.

Further technical improvements could include materialised workload views, gRPC-based service communication, more advanced observability dashboards, automated field-level audit trails, accessibility testing and improved backup and disaster recovery procedures. These steps would help move the prototype toward production-level deployment.

APPENDICES

Appendix A Integrated interface specification

The following summary records the main integrated interface contracts used by the common solution. It is included to verify how the components exchange data through the gateway and shared database.

Table 7 . contract,input,output and notes

Contract	Input	Output	Notes
Recommendation request	Research title, abstract, keywords and requested number of results	Ranked supervisor cards with scores, workload labels and request metadata	Invoked through the backend gateway.
Gap analysis request	Research idea text and related metadata	Classification, confidence and supporting evidence	Supports early-stage idea validation.
Integrity attestation request	Submission metadata and document hash	Attestation status and integrity evidence	Can be processed asynchronously.
Logbook summary request	Student activity text and review context	Concise activity summary and optional feedback support	Original logbook remains available for verification.

Appendix B Team contribution declaration

The team confirms that each member was responsible for an individual component and also contributed to shared integration, testing and report preparation. The table below summarises the agreed contribution record.

Table 8 . contribution

Member	Confirmed contribution
S.M.D.M. Bandara	Semantic analysis and gap detection component, integration of idea validation output and related documentation.
K.L. Widanagama	Supervisor recommendation and collaboration component, workload-aware request workflow

	and integration with assignment state.
Darshan R	Integrity governance and blockchain ledger component, submission integrity workflow and attestation support.
Wijesinghe L.P.P.	Collaborative workspace and NLP supervisor assistant component, logbook management and summarisation support.
All members	Shared project planning, API coordination, integration testing, validation walkthroughs and preparation of the common integrated solution report.

Appendix C Academic integrity and AI use disclosure

Generative AI assistance was used for limited support purposes in preparing this common report, including paragraph refinement, structure planning, document formatting support and image-generation assistance for explanatory diagrams. The team reviewed, edited and verified the final content before inclusion. The final responsibility for technical accuracy, citations, implementation evidence and conclusions remains with the research team.